INCH-POUND
MIL-PRF-19565D
w/AMENDMENT 1
6 January 2021
SUPERSEDING
MIL-PRF-19565D
10 April 2019

PERFORMANCE SPECIFICATION

COATING COMPOUND, THERMAL INSULATION, FIRE- AND WATER-RESISTANT, VAPOR-BARRIER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers type II fire- and water-resistant thermal insulation vapor-barrier coating compound for interior use in insulated refrigerant and chilled water lines.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2819 - Insulation Block, Thermal

MIL-C-20079 - Cloth, Glass; Tape, Textile Glass; and Thread, Glass and Wire-Reinforced Glass

MIL-DTL-24607 - Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin, Semigloss

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8030

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AK-DPC-010/078-1 - Composite Materials, Surface Ships, Topside Structural and Other Topside Applications – Fire Performance Requirements

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

DELETED

ASTM D92 - Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester

ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials

ASTM E96/E96M - Standard Test Methods for Water Vapor Transmission of Materials

ASTM F1249 - Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The coating compound furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

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3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Stability.

3.3.1 Storage stability. The compound shall not liver, settle, or become unusable when stored for a period of 1 year in an airtight container (see 4.4.1.1) and shall meet the original manufacturer's specifications after that year.

3.3.2 Freeze thaw stability. The compound shall not visually change, separate, or shrink when tested as specified in 4.4.1.2.

3.4 Water vapor permeance. The water vapor permeance of the compound shall be not greater than 0.05 perm (see 4.4.2).

3.5 Flash point. The flash point of the compound shall be not less than 80 °F (see 4.4.3).

3.6 Flexibility. The compound shall not chip, scale, crack, or otherwise disbond from the surface of a metal strip when tested as specified in 4.4.4. Slight surface cracks and edge cracking shall be disregarded, but cracks that expose bare metal shall be considered as failure of the compound to pass the test.

3.7 Water resistance. The compound shall not show evidence of leaching, disintegration, or loss of adhesion when tested as specified in 4.4.5.

3.8 Protection against corrosion. The compound shall not have more than a 0.25-inch creep of rust or corrosion under the coating from either side of the scribe line when tested as specified in 4.4.6.

3.9 Paint compatibility. Prior to immersion for the tape test (see 4.4.7) and after the compound has cured, it shall not cause discoloration or bleeding through a coat of paint conforming to MIL-DTL-24607. The compound shall only leave trace evidences of paint when tested for adhesion as specified in 4.4.7. Neither the paint nor the compound shall soften or blister on the specimen.

3.10 Fire performance.

3.10.1 Surface flammability. The compound shall be fire resistant, shall have a flame spread index not greater than 25, and shall not exhibit melting, dripping, or flaming droplets when tested as specified in 4.4.8.1.1.

3.10.2 Smoke density. The compound shall have a smoke density not greater than 50 and shall not exhibit melting, dripping, or flaming droplets when tested as specified in 4.4.8.1.2.

3.10.3 Fire gas toxicity. The compound shall meet the requirements for fire gas toxicity in accordance with T9070-AK-DPC-010/078-1, except the limit for carbon monoxide shall be 600 parts per million, when tested as specified in 4.4.8.1.3.

3.11 Toxicity. When evaluated in accordance with 4.4.9, the compound shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.4.9 and 6.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the tests of 4.4.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.4 and the tests of 4.4.3 and 4.4.4.

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4.3.1 Lot. For the purposes of inspection, a lot shall consist of all compound from one production batch offered for delivery at one time.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-1916 or ASQ/ANSI Z1.4 at inspection level I and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.3 Sampling for tests. Two containers shall be randomly selected from each lot. Two 1-quart samples shall be taken from each of the containers for the tests specified in 4.3.

4.3.4 Examination of filled containers. Each sample-filled container selected as specified in 4.3.2 shall be examined to verify compliance with this specification. Any container in the sample having one or more defects or that is under required fill shall be rejected. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan in accordance with MIL-STD-1916 or ASQ/ANSI Z1.4, the entire lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 Stability.

4.4.1.1 Storage stability. The compound shall be stored for a period of 1 year in an airtight container at room temperature (71.6±3.6 °F). At the end of this period, the compound shall be visually examined to determine conformance with 3.3.1 and shown to meet the original manufacturer's specifications.

4.4.1.2 Freeze thaw stability. The compound shall be subjected to a temperature of 16±2 °F for 16 hours ±1 minute and returned to room temperature (71.6±3.6 °F) before testing for stability.

4.4.2 Water vapor permeability.

4.4.2.1 Apparatus. The apparatus shall be in accordance with ASTM E96/E96M or ASTM F1249.

4.4.2.2 Procedure. The procedure shall be in accordance with procedure A (Desiccant Method) of ASTM E96/E96M or ASTM F1249.

4.4.2.3 Test specimens. The supporting membrane of the test specimen shall be of any porous, semi-rigid material, such as the glass-cloth facing of fibrous glass hull insulation board. A 0.0625-inch-thick wet coat of the compound shall be applied to the supporting membrane with a doctor blade. The membrane shall be of a size to yield four specimens. The coated specimens shall be air-dried at room temperature (71.6±3.6 °F) for 72 hours prior to assembly.

4.4.3 Flash point. The flash point shall be determined in accordance with ASTM D92.

4.4.4 Flexibility.

4.4.4.1 Apparatus. The apparatus shall consist of the following:

a. A tin plate panel, 8 by 18 inches, cleaned and dried using a 50:50 mixture of petroleum naphtha and 2-ethoxy ethanol.

b. A film applicator with a clearance of 0.0625 inch and a gap of sufficient size to make a film 3 to 6 inches wide.

c. A steel mandrel, 0.25 inch in diameter and 6 inches long, supported at each end.

4.4.4.2 Procedure. The 8- by 18-inch tin panel (0.013 inches) shall be placed on a flat surface. Using the film applicator, a film of the compound 18 inches long and 3 to 6 inches wide shall be applied on the panel. The coated panel shall be air dried for no less than 72 hours and up to 2 weeks at 73±1 °F and 50±2 percent relative humidity. Four 1½- by 4-inch strips shall be cut from the coated portion of the panel. Two of the strips shall be placed in an electric air-circulating oven and baked at 158±2 °F for 48 hours. The other two strips shall be placed in tap water at a temperature of 75 °F for 48 hours. The strips shall be removed from the oven and the water allowed to come to room temperature (71.6±3.6 °F), then the strips shall be bent rapidly over the surface of the mandrel and examined for cracks at the bend. Failure of the strip to meet the requirements in 3.6 shall be cause for rejection.

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4.4.5 Water resistance. The test specimen, prepared as specified in 4.4.4.2, shall be observed for evidence of leaching, disintegration, or loss of adhesion.

4.4.6 Corrosion protection.

4.4.6.1 Apparatus. The apparatus used shall be in accordance with ASTM B117.

4.4.6.2 Procedure. Before testing, a line shall be scribed through the coating to expose the underlying metal surface. A 0.125-inch-thick wet coating of the compound shall be applied to unprimed cold-rolled steel (SAE-AISI 1008/1010) panels with a smooth finish, 0.02 inch thick, and air-dried for 30 days. The coated panels shall be subjected to salt spray for 30 days and then examined for signs of rusting or corrosion.

4.4.7 Paint compatibility. A block of thermal insulation in accordance with class 2 of MIL-PRF-2819, 6±0.1 by 4±0.1 by 2±0.1 inches thick, shall be covered with a continuous piece of fiber glass cloth conforming to MIL-C-20079, cut so that there are a minimum number of seams. The cloth shall be secured tightly to the surface of the block. A coat of type II compound shall be applied according to the manufacturer-recommended thickness and allowed to dry for 72 hours. A coat of enamel in accordance with MIL-DTL-24607 shall then be applied to the specimen. After painting, coating shall be cured for no more than 2 weeks before further testing. All coatings shall be applied at room temperature (71.6±3.6 °F). Adhesion shall be determined in accordance with method 6301 (Adhesion [Wet] Tape Test) of FED-STD-141.

4.4.8 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.4).

4.4.8.1 Fire resistance.

4.4.8.1.1 Surface flammability. Test samples shall be prepared by coating a 0.25-inch-thick steel substrate or inorganic cement board with the vapor-barrier compound at the maximum thickness anticipated for use and allowing time for the coating to dry (cure). The samples shall then be tested in accordance with ASTM E84.

4.4.8.1.2 Smoke density. Test samples shall be prepared by coating a 0.25-inch-thick steel substrate or inorganic cement board with the vapor-barrier compound at the maximum thickness anticipated for use and allowing time for the coating to dry (cure). The samples shall then be tested in accordance with ASTM E84.

4.4.8.1.3 Fire gas toxicity. Maximum toxic gas concentrations shall be measured and recorded in accordance with T9070-AK-DPC-010/078-1. The coating shall be allowed to dry (cure) before the gas concentrations are measured.

4.4.9 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.11 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the compound using data provided by the manufacturer/distributor to the NMCPHC (see 3.11 and 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The compound is intended for use as a vapor-barrier coating over thermal insulation on U.S. Navy ships. This product must survive in a maritime environment in excess of commercial standards and operate at higher levels of reliability to meet the Navy's mission.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. AQL of 2.5 percent (see 4.3.2).

c. Additional fire testing provisions (see 4.4.8 and 6.4).

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 19565 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.6 Subject term (key word) listing.

Barrier, plastic

Damp-proofing

Moisture prevention, insulation

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – CE
Navy – SH

Review activities:
Army – CR4
Navy – AS, YD

Preparing activity:
Navy − SH
(Project 8030-2020-004)

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